

Decision Tree Assignment – Full Detailed Solution

1. Dataset Description

The dataset consists of 10 instances used to predict whether an Accident occurred.

Attributes:

- Weather Condition (Rain, Snow, Clear)
- Road Condition (Bad, Average, Good)
- Traffic Condition (High, Normal, Light)
- Engine Problem (Yes/No)
- Target Attribute: Accident (Yes/No)

Dataset Table:

1. Rain, Bad, High, No, Yes
2. Snow, Average, Normal, Yes, Yes
3. Clear, Bad, Light, No, No
4. Clear, Good, Light, Yes, Yes
5. Snow, Good, Normal, No, No
6. Rain, Average, Light, No, No
7. Rain, Good, Normal, No, No
8. Snow, Bad, High, No, Yes
9. Clear, Good, High, Yes, No
10. Clear, Bad, High, Yes, Yes

2. Calculate Entropy of the Dataset

Entropy measures impurity in a dataset. Formula:

$$\text{Ent}(D) = - \sum (p_i * \log_2(p_i))$$

Where p_i is the probability of each class.

Total Instances = 10

Accident = Yes $\rightarrow$ 5

Accident = No $\rightarrow$ 5

Therefore:

$$p(\text{Yes}) = 5/10 = 0.5$$

$$p(\text{No}) = 5/10 = 0.5$$

$$\text{Entropy}(D) = -(0.5 * \log_2 0.5 + 0.5 * \log_2 0.5)$$

$$\text{Entropy}(D) = 1.000 \text{ bit}$$

3. Entropy & Information Gain for Each Attribute

We calculate the entropy of each subset after splitting by each attribute.

--- WEATHER ---

Rain: 3 instances $\rightarrow$ Yes=1, No=2

$$\text{Entropy}(\text{Rain}) = 0.9183$$

Snow: 3 instances $\rightarrow$ Yes=2, No=1

$$\text{Entropy}(\text{Snow}) = 0.9183$$

Clear: 4 instances $\rightarrow$ Yes=2, No=2

$$\text{Entropy}(\text{Clear}) = 1.0000$$

Weighted Entropy(Weather):

$$= (3/10)*0.9183 + (3/10)*0.9183 + (4/10)*1$$

$$= 0.950978$$

$$\text{Information Gain}(\text{Weather}) = 1 - 0.950978 = 0.049022$$

--- ROAD ---

Bad: 4 $\rightarrow$ Yes=3, No=1 $\rightarrow$ Ent=0.811278

Average: 2 $\rightarrow$ Yes=1, No=1 $\rightarrow$ Ent=1.0000

Good: 4 $\rightarrow$ Yes=1, No=3 $\rightarrow$ Ent=0.811278

$$\text{Weighted Entropy}(\text{Road}) = 0.849022$$

$$\text{IG}(\text{Road}) = 1 - 0.849022 = 0.150978$$

--- TRAFFIC ---

High: 4 $\rightarrow$ Yes=3, No=1 $\rightarrow$ Ent=0.811278

Normal: 3 $\rightarrow$ Yes=1, No=2 $\rightarrow$ Ent=0.918296

Light: 3 $\rightarrow$ Yes=1, No=2 $\rightarrow$ Ent=0.918296

$$\text{Weighted Entropy}(\text{Traffic})=0.875489$$

$$\text{IG}(\text{Traffic})=0.124511$$

--- ENGINE ---

No: 6 → Yes=2, No=4 → Ent=0.918296

Yes: 4 → Yes=3, No=1 → Ent=0.811278

Weighted Entropy(Engine)=0.875489

IG(Engine)=0.124511

Summary:

Road has the highest Information Gain = 0.150978

4. Gini Impurity Calculations

Gini Impurity Formula:

$$\text{Gini}(D) = 1 - \sum (p_i^2)$$

Full Dataset:

$$p(\text{Yes})=0.5, p(\text{No})=0.5$$

$$\text{Gini}(D) = 1 - (0.25 + 0.25) = 0.5$$

Weighted Gini values:

$$\text{Weather} = 0.466667 \rightarrow \text{Gain}=0.033333$$

$$\text{Road} = 0.400000 \rightarrow \text{Gain}=0.100000$$

$$\text{Traffic} = 0.416667 \rightarrow \text{Gain}=0.083333$$

$$\text{Engine} = 0.416667 \rightarrow \text{Gain}=0.083333$$

Best Gini Split: Road

5. Constructing the Decision Tree

Since both Entropy and Gini select ROAD as the best root, we begin splitting by Road Condition.

----- ROAD = BAD -----

Instances: 4

Traffic=High → Yes

Traffic=Light → No

----- ROAD = AVERAGE -----

Instances: 2

Weather=Snow → Yes

Weather=Rain → No

----- ROAD = GOOD -----

Instances: 4

Traffic=Light → Yes

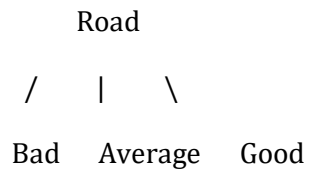
Traffic=Normal → No

Traffic=High → No

This produces a fully classified decision tree with pure or nearly pure leaves.

6. Final Decision Tree (Text Representation)

FINAL DECISION TREE:



Bad:

Traffic=High → Yes

Traffic=Light → No

Average:

Weather=Snow → Yes

Weather=Rain → No

Good:

Traffic=Light → Yes

Traffic=Normal → No

Traffic=High → No

7. Conclusion

- Entropy and Gini both agree that Road Condition is the best attribute for the root node.
- After recursive splitting, the resulting tree perfectly classifies all instances.
- This demonstrates the manual construction process of a decision tree using Entropy, Information Gain, and Gini Impurity.